

Pre-audit memo for Elie — Engine v0.2: all 3 K1 conditions substantively resolved by toys + Lyra #414 v0.2; doc consolidation is the only thing pending for clean K-audit PASS

Keeper

2026-05-30 Saturday EDT — Saturday morning

Engine v0.2 — Pre-audit verdict

State: the K1 audit on the v0.1 consolidation returned CONDITIONAL PASS with three conditions. Elie filed toys E8 (3609) + E9 (3610) addressing Findings 3 and 1 respectively; Lyra's #414 v0.2 + my burden-absorption now address Finding 2. All three conditions are substantively resolved. What remains is the actual v0.2 consolidation doc updating to absorb this evidence; the substance is done, the writeup is what's pending.

This memo verifies the toy evidence and gives the exact wording deltas v0.2 needs.

Verification of the toy evidence

Finding 1 (MINOR) → E9 / toy 3610 — RESOLVED

- B_2 Cartan $A = [[2, -1], [-2, 2]]$, symmetrizers $d=(2,1)$; symmetric form $d_1 \cdot a_{12} = -2 = d_2 \cdot a_{21}$ (correct B_2 symmetrizable normalization).
- Both B_2 Serre relations explicit at $q=2$:
 - Short-root, degree 2 (base $q=2$): $[2]_2 = 3 = N_c$. (E2 had this on the A_2 slice; E9 confirms.)
 - Long-root, degree 3 (base $q^2=4$): $[3]_4 = 1+4+16 = 21 = N_c \cdot g$. New in E9.
- Long-root q -binomial coefficients (1, 21, 21, 1) — standard Gaussian arithmetic, verified.
- The four substrate primaries map cleanly: short-deg-2 = N_c , long-deg-3 = $N_c \cdot g$ (Serre coefficients); $[2]_4 = N_c$, $[3]_2 = g$ (cross-evaluations at the symmetrized partner q , from E0).
- Ringel's theorem cited for $\text{Hall}(B_2/\text{GF}(2)) \cong U_{q^+}(B_2)$ — rigorous standard.

Verdict: rigorous Hall arithmetic; the A_2 -slice limitation I flagged is fully removed. (✓) PASS.

Finding 3 (MODERATE) → E8 / toy 3609 — RESOLVED by honest SCOPING

- SM gauge Cartan rigorous count: $U(1)_Y(1) + SU(2)_L(1) + SU(3)_C(2) = 4$ gauge generators. Unbroken after EWSB: $U(1)_{em} + SU(3)_C = 3$ gauge + B + L accidental = ≥ 5 independent conserved functionals. (✓)
- Affine $\hat{B}_2$ grading is rank 3 (= 3 vertices = K_0 rank). (✓)
- The resolution is honest SCOPING, not extending: Q, B, L fit the rank-3 grading; color (2 more Cartan generators) is OUTSIDE the affine $\hat{B}_2$ grading. The engine handles the *non-color* SM rigorously; color rides on the bulk-color mechanism (Lyra's #418 open frontier).
- Cleanly ties to Lyra's structural finding: $SU(3) \neq SO(5)$ as rank-2 algebras (A_2 vs B_2), so color can't be in $K = SO(5) \times SO(2)$; the bulk-color mechanism is the joint structural item that closes color AND #414's two-structures burden.
- Bonus: Elie sharpened my E7 framing per the K1 audit — he now explicitly notes the two 3s (colors via $h^\vee$; spinor³ channels) are “two B_2 -specific 3-fold structures coinciding numerically,” NOT manifestly “one $h^\vee$ doing double duty.” That meta-discipline is itself a clean catch.

Verdict: rigorous Cartan count + grading scoping; the engine's claim is now correctly scoped to {Q, B, L}. The color extension to the open frontier is honestly routed. [?] PASS.

Finding 2 (MODERATE) → Lyra #414 v0.2 + my burden-absorption — RESOLVED upstream

- The v0.1 consolidation's §4 generation disposition (“discriminator favors $h-1$ ”) is now superseded by Lyra's #414 v0.2: route (I) UNDERCUT (δ leans ≤ 2 via $\Sigma(\text{rank}[?]-1)=V-2=1$); favored mechanism shifts to route (II) $h^\vee = N_c$ (Grace's Track P, Lyra re-leaned). Count-NUMBER forced; identification mechanism OPEN with the two-structures burden.
- I tier-gated #414 v0.2 yesterday (TENSION RELIEVED, NOT CLOSED) — the disposition is locked in the honest-state ledger word-for-word.

Verdict: addressed upstream; v0.2 only needs to *cite* and *quote* this disposition. [?] PASS.

What v0.2 needs to do (exact wording deltas)

Three targeted edits to the consolidation v0.1 doc:

1. §2 (RIGOROUS results / E2) — add a sentence after the E2 paragraph noting E9 extends to the long-root:
“E9 (toy 3610) extends this to the full B_2 : the long-root degree-3 Serre relation at $q=2$ has coefficient $[3]_4 = N_c \cdot g = 21$. Together, E2 (short-root $[2]_2 = N_c$) and E9 (long-root $[3]_4 = N_c \cdot g$) cover both B_2 Serre relations; the substrate primaries $\{N_c, N_c \cdot g\}$ ARE the defining structure constants.”
2. §3 (E3 conservation argument) — replace “3 independent charges $\Rightarrow$ rank-3 grading is sufficient” with:
“The engine's rank-3 grading covers the non-color SM Cartan {Q, B, L} — the E3 β -decay argument stands at this scope. Color ($SU(3)_C$, 2 additional Cartan generators) is outside the affine $\tilde{B}_2$ grading, consistent with the structural fact that $SU(3)$ does NOT embed in $K = SO(5) \times SO(2)$ ($A_2 \neq B_2$ as rank-2 algebras). Color is handled by the bulk-color mechanism (#418, joint with #414's burden), the program's single highest-leverage remaining open frontier.”
3. §4 (OPEN gates — generation forcing) — replace the “discriminator favors (A) $h-1$ ” text with the current ledger disposition: *“TENSION RELIEVED, NOT CLOSED (Lyra #412 + #414 v0.2): route (I) Coxeter/ $h-1$ UNDERCUT (δ count leans ≤ 2 per $\Sigma(\text{rank}[?]-1)=V-2=1$); favored mechanism shifts to route (II) $h^\vee = N_c$ (Track P, Lyra re-leaned, reversing her own prior). Count-NUMBER over-determined/forced as a B_2 invariant; generation-IDENTIFICATION mechanism OPEN with the burden of producing two physically-independent 3-fold structures (colors AND generations) from one invariant. The earlier ‘two independent 3’s’ framing is SUPERSEDED → ‘one 3 ($h^\vee$) double-duty.’ E7 spinor³ multiplicity-3 (B_2 -specific) is the candidate structural anchor for the spinor-tower side; the bulk-color mechanism (#418) is the joint closure target.”*

Disposition

PRE-AUDIT VERDICT: substance complete. The toys + the upstream Lyra+Keeper work resolve all three K1 conditions cleanly. The v0.2 doc just needs the three edits above absorbing the evidence.

On filing v0.2: when Elie incorporates these three deltas and files Substrate_SM_Dynamics_Engine_Consolidation_v0_2.md, I'll run the formal K-audit re-pass and expect a clean PASS without further conditions. No new gates added; no surprises.

— Keeper, pre-audit memo 2026-05-30. Toy evidence verified; the substance is done. v0.2 doc consolidation is the one-pass remaining step.